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LabSheet:02

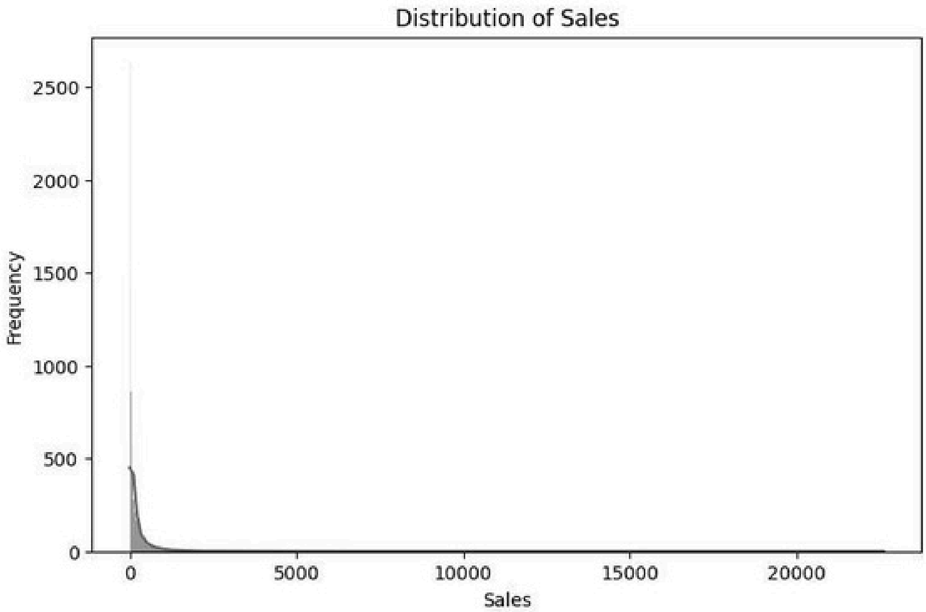
```
#1. Import libraries and load dataset
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv("superstore.csv")

# Display first 5 rows
print(df.head())
```

```
  Row ID        Order ID  Order Date   Ship Date       Ship Mode Customer ID  \
0      1  CA-2017-152156   11/8/2017  11/11/2017    Second Class    CG-12520
1      2  CA-2017-152156   11/8/2017  11/11/2017    Second Class    CG-12520
2      3  CA-2017-138688   6/12/2017   6/16/2017    Second Class    DV-13045
3      4  US-2016-108966  10/11/2016  10/18/2016  Standard Class    SO-20335
4      5  US-2016-108966  10/11/2016  10/18/2016  Standard Class    SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-Category  \
0     42420.0   South  FUR-BO-10001798        Furniture    Bookcases
1     42420.0   South  FUR-CH-10000454        Furniture       Chairs
2     90036.0    West  OFF-LA-10000240  Office Supplies       Labels
3     33311.0   South  FUR-TA-10000577        Furniture       Tables
4     33311.0   South  OFF-ST-10000760  Office Supplies      Storage

                                        Product Name     Sales  Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600       2.0
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400       3.0
2  Self-Adhesive Address Labels for Typewriters b...   14.6200       2.0
3      Bretford CR4500 Series Slim Rectangular Table  957.5775       5.0
4                     Eldon Fold 'N Roll Cart System   22.3680       2.0

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]
```

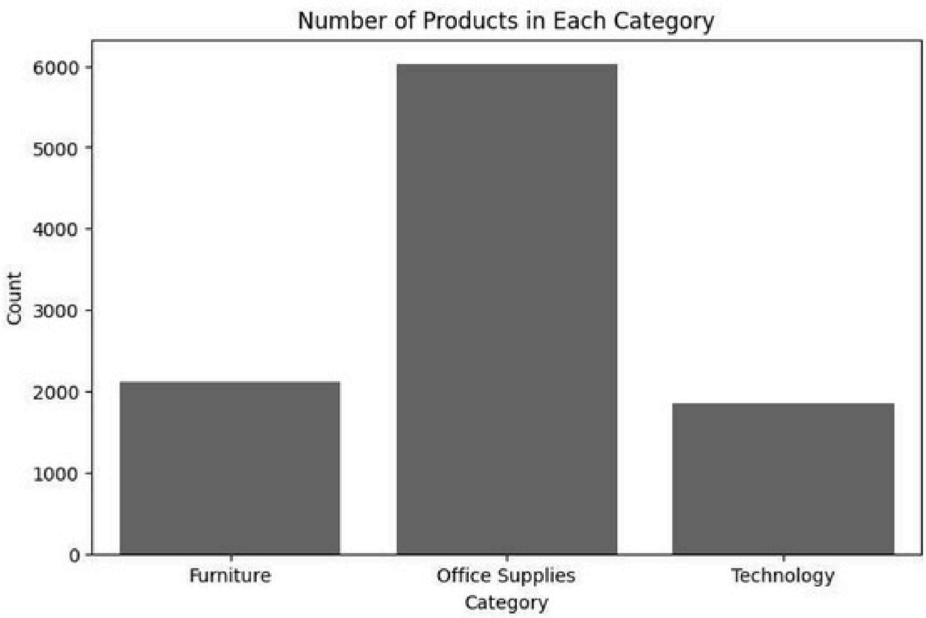
```
#2. Explore the dataset
# Number of rows and columns
print("Shape of dataset:", df.shape)

# Column names
print("\nColumn Names:")
print(df.columns)

# Information about dataset
print("\nDataset Information:")
df.info()

# Statistical summary
print("\nStatistical Summary:")
print(df.describe())

# Check unique values in categorical columns
print("\nCategory Counts:")
print(df["Category"].value_counts())
```

```
Shape of dataset: (10800, 21)
```

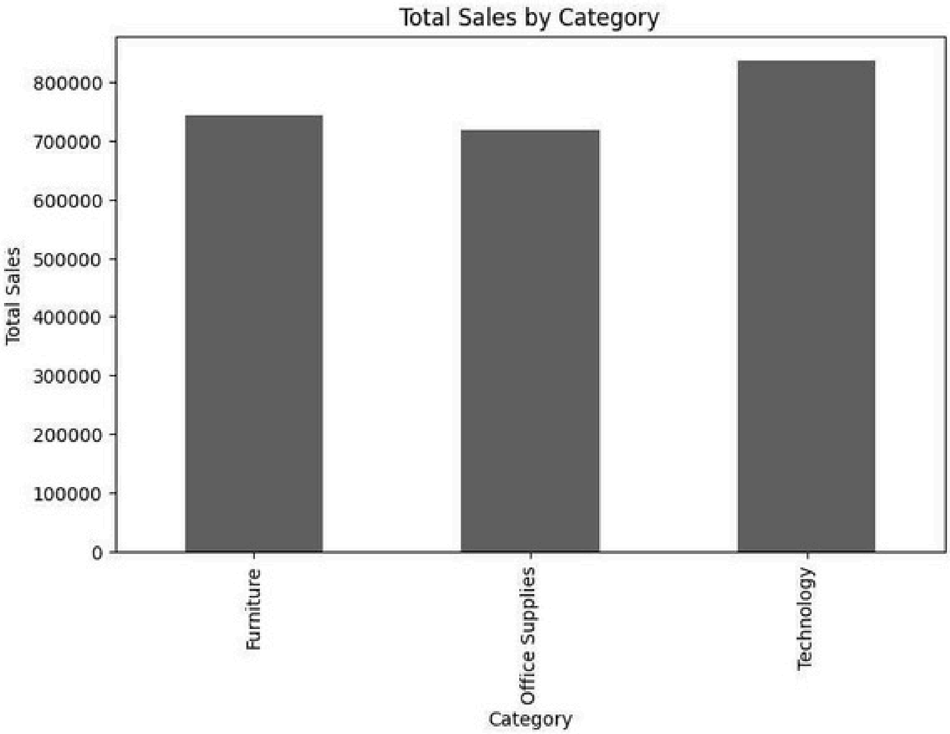
```
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")

plt.show()
```




```
#5. Categorical variable – Count Plot
plt.figure(figsize=(8, 5))

sns.countplot(x="Category", data=df)

plt.title("Number of Products in Each Category")
plt.xlabel("Category")
plt.ylabel("Count")

plt.show()
```



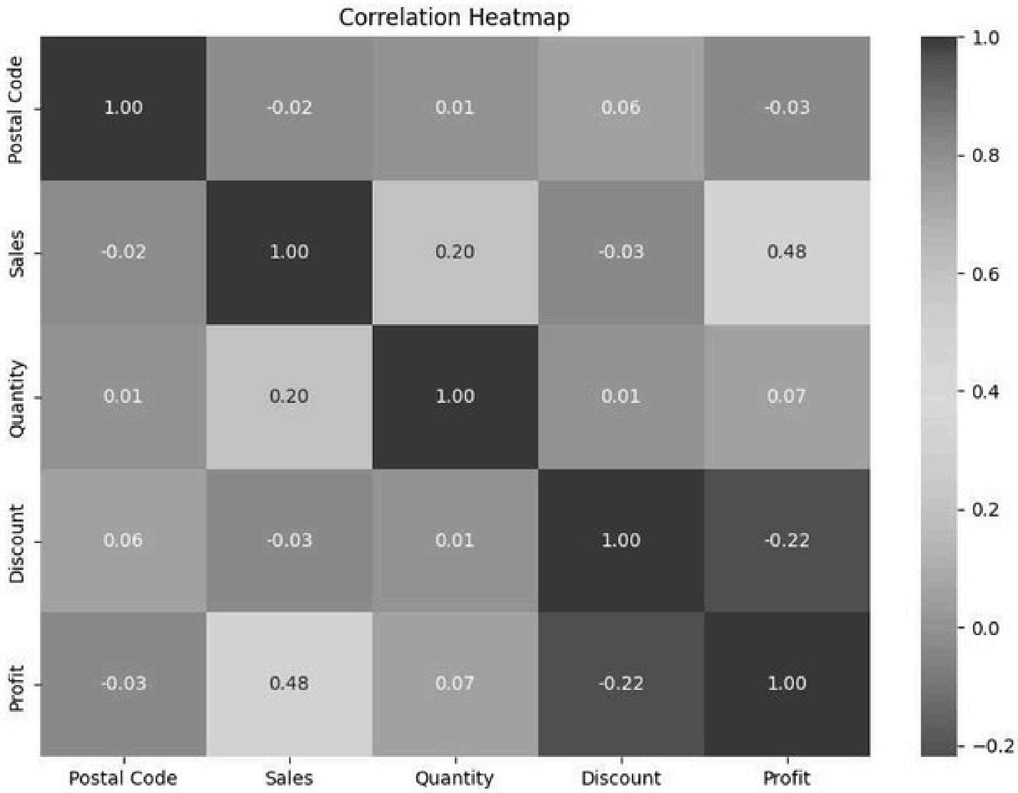

```
#6. Bar chart – Sales by Category
category_sales = df.groupby("Category")["Sales"].sum()

plt.figure(figsize=(8, 5))

category_sales.plot(kind="bar")

plt.title("Total Sales by Category")
plt.xlabel("Category")
plt.ylabel("Total Sales")

plt.show()
```



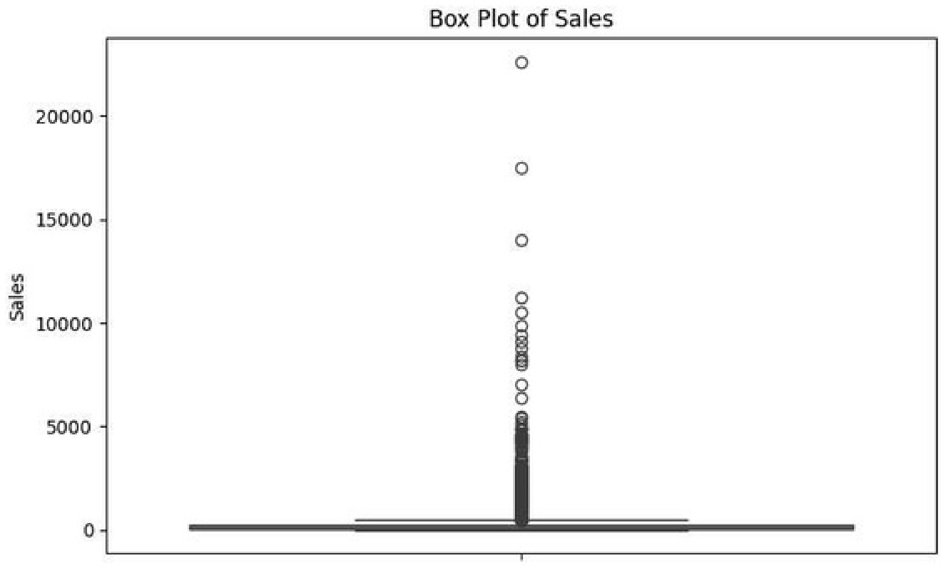

```
#7. Correlation Matrix
# Select numerical columns
numeric_data = df.select_dtypes(include=np.number)

# Correlation matrix
correlation = numeric_data.corr()

print(correlation)
```

```
             Postal Code     Sales  Quantity  Discount    Profit
Postal Code     1.000000 -0.022346  0.013376  0.057243 -0.028751
Sales          -0.022346  1.000000  0.200795 -0.028190  0.479064
Quantity        0.013376  0.200795  1.000000  0.008623  0.066253
Discount        0.057243 -0.028190  0.008623  1.000000 -0.219487
Profit         -0.028751  0.479064  0.066253 -0.219487  1.000000
```

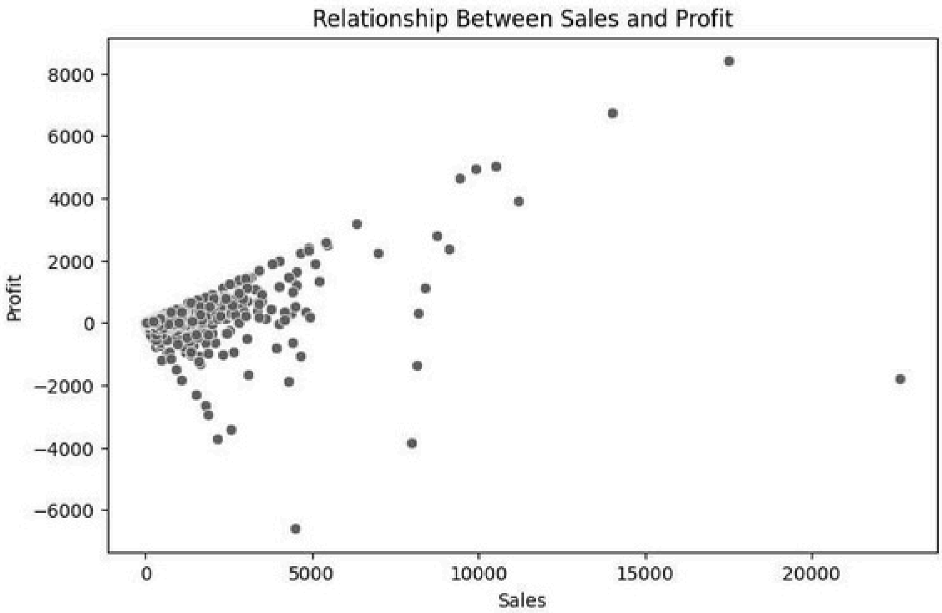
```
#8. Correlation Heatmap
plt.figure(figsize=(10, 7))

sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

plt.title("Correlation Heatmap")

plt.show()
```



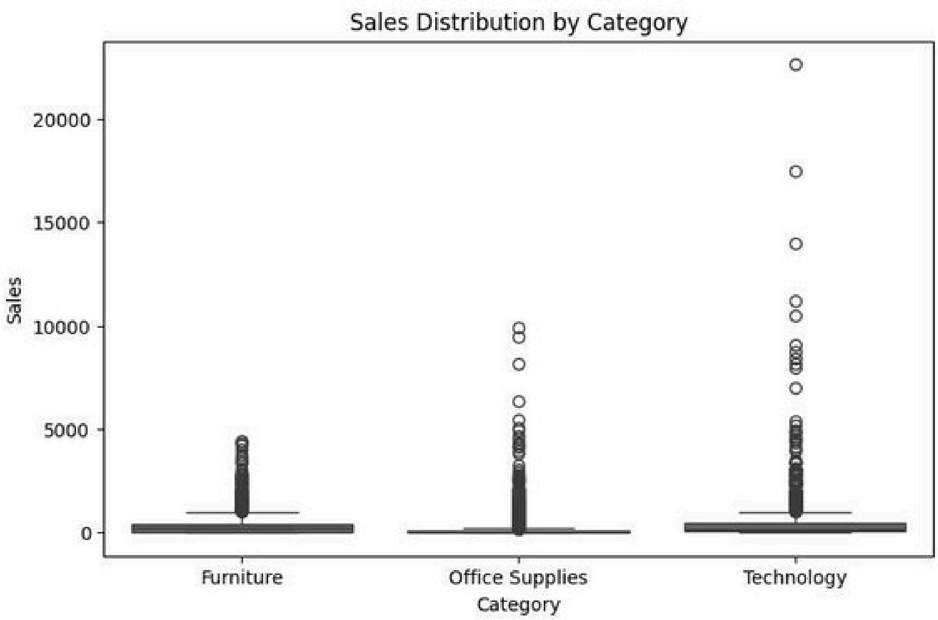

```
#9. Box Plot — Detect Outliers
plt.figure(figsize=(8, 5))

sns.boxplot(y=df["Sales"])

plt.title("Box Plot of Sales")
plt.ylabel("Sales")

plt.show()
```



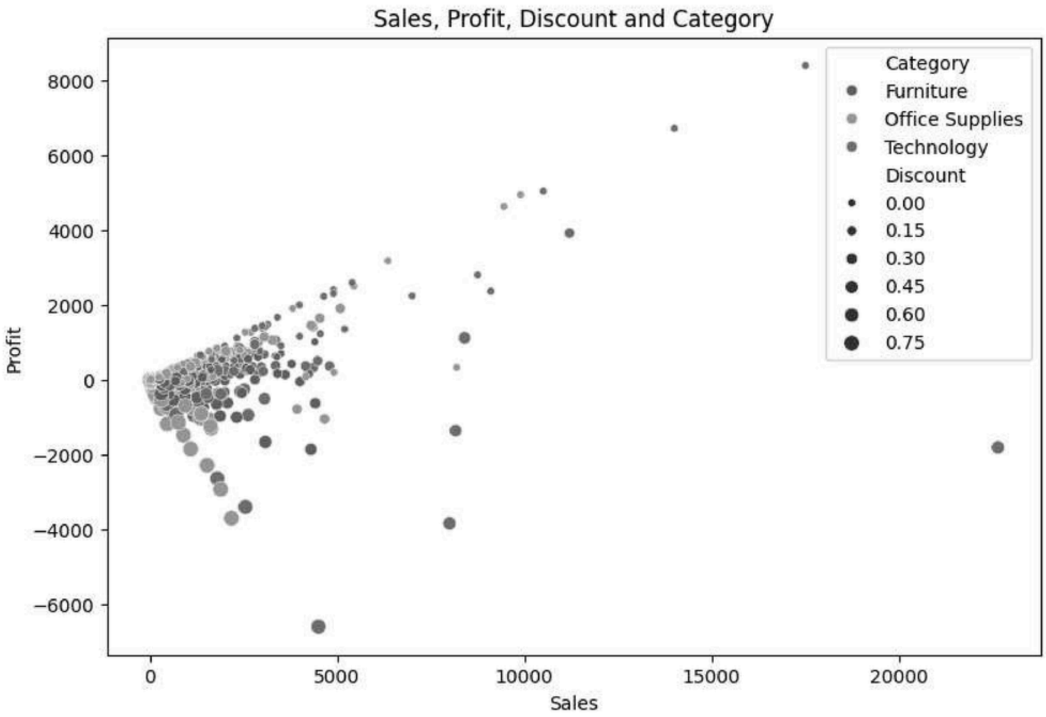
```python
#10. Scatter Plot — Sales vs Profit
plt.figure(figsize=(8, 5))

sns.scatterplot(
    x="Sales",
    y="Profit",
    data=df
)

plt.title("Relationship Between Sales and Profit")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()
```




```python
#11. Bivariate Analysis — Sales by Category
plt.figure(figsize=(8, 5))

sns.boxplot(
    x="Category",
    y="Sales",
    data=df
)

plt.title("Sales Distribution by Category")
plt.xlabel("Category")
plt.ylabel("Sales")

plt.show()
```



```
#12. Multivariate Analysis
plt.figure(figsize=(9, 6))

sns.scatterplot(
    x="Sales",
    y="Profit",
    hue="Category",
    size="Discount",
    data=df
)

plt.title("Sales, Profit, Discount and Category")
plt.xlabel("Sales")
plt.ylabel("Profit")

plt.show()
```




```
#13. Useful business insights
# Category with highest sales
highest_sales_category = (
    df.groupby("Category")["Sales"]
    .sum()
    .idxmax()
)

print("Category with highest sales:", highest_sales_category)


# Category with highest profit
highest_profit_category = (
    df.groupby("Category")["Profit"]
    .sum()
    .idxmax()
)

print("Category with highest profit:", highest_profit_category)


# Most common category
most_common_category = df["Category"].value_counts().idxmax()

print("Most common category:", most_common_category)
```

```
Category with highest sales: Technology
Category with highest profit: Technology
Most common category: Office Supplies
```

Final observation:

The dataset contains both numerical and categorical variables.

```
Column Names:
Index(['Row ID', 'Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City', 'State',
       'Postal Code', 'Region', 'Product ID', 'Category', 'Sub-Category',
       'Product Name', 'Sales', 'Quantity', 'Discount', 'Profit'],
      dtype='object')

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10800 entries, 0 to 10799
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype  
---  ------         --------------  -----  
 0   Row ID         10800 non-null  object 
 1   Order ID       10800 non-null  object 
 2   Order Date     9994 non-null   object 
 3   Ship Date      9994 non-null   object 
 4   Ship Mode      9994 non-null   object 
 5   Customer ID    9994 non-null   object 
 6   Customer Name  9994 non-null   object 
 7   Segment        9994 non-null   object 
 8   Country        9994 non-null   object 
 9   City           9994 non-null   object 
 10  State          9994 non-null   object 
 11  Postal Code    9983 non-null   float64
 12  Region         9994 non-null   object 
 13  Product ID     9994 non-null   object 
 14  Category       9994 non-null   object 
 15  Sub-Category   9994 non-null   object 
 16  Product Name   9994 non-null   object 
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   float64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(5), object(16)
memory usage: 1.7+ MB

Statistical Summary:
        Postal Code         Sales     Quantity     Discount       Profit
count   9983.000000   9994.000000  9994.000000  9994.000000  9994.000000
mean   55245.233297    229.858001     3.789574     0.156203    28.656896
std    32038.715955    623.245101     2.225110     0.206452   234.260108
min     1040.000000      0.444000     1.000000     0.000000 -6599.978000
25%    23223.000000     17.280000     2.000000     0.000000     1.728750
50%    57103.000000     54.490000     3.000000     0.200000     8.666500
75%    90008.000000    209.940000     5.000000     0.200000    29.364000
max    99301.000000  22638.480000    14.000000     0.800000  8399.976000

Category Counts:
Category
Office Supplies    6026
Furniture          2121
Technology         1847
Name: count, dtype: int64
```

```
#3. Check missing values
print("Missing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Row ID             0
Order ID           0
Order Date       806
Ship Date        806
Ship Mode        806
Customer ID      806
Customer Name    806
Segment          806
Country          806
City             806
State            806
Postal Code      817
Region           806
Product ID       806
Category         806
Sub-Category     806
Product Name     806
Sales            806
Quantity         806
Discount         806
Profit           806
dtype: int64
```

```
#4. Univariate Analysis – numerical variable
plt.figure(figsize=(8, 5))

sns.histplot(df["Sales"], kde=True)
```

- The dataset contains both numerical and categorical variables.
- The histogram shows the distribution of sales.
- The count plot shows the frequency of different product categories.
- The box plot helps identify unusually high or low sales values.
- The scatter plot shows the relationship between sales and profit.
- The correlation heatmap shows the strength of relationships between numerical variables.
- Category-wise analysis helps identify which category contributes most to sales and profit.

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LabSheet:02

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Q1. What is Exploratory Data Analysis (EDA), and why is it performed before machine learning?

Answer: Exploratory Data Analysis (EDA) is the process of examining and analyzing a dataset using statistical methods and visualizations to understand its structure, patterns, relationships, and anomalies.

EDA is performed before machine learning because it helps to:

- Understand the dataset and its variables.
- Identify missing or incorrect values.
- Detect outliers and anomalies.
- Understand the distribution of data.
- Find relationships between features.
- Detect highly correlated or irrelevant features.
- Decide which preprocessing techniques are required.
- Improve the quality and performance of machine learning models.

Q2. Differentiate between univariate, bivariate, and multivariate analysis with suitable examples.

Answer:

Analysis	Meaning	Example
Univariate	Analysis of a single variable	Analyzing the distribution of Sales using a histogram
Bivariate	Analysis of two variables	Studying the relationship between Sales and Profit using a scatter plot
Multivariate	Analysis of three or more variables	Studying how Sales, Profit, Discount, and Category are related

Example: If we analyze only **Sales**, it is univariate analysis. If we analyze **Sales and Profit**, it is bivariate analysis. If we analyze **Sales, Profit, Discount, and Category** together, it is multivariate analysis.

Q3. What insights can be obtained from a correlation heatmap?

Answer: A correlation heatmap represents the correlation between numerical variables using colors and correlation values.

It can help us:

- Identify strongly positively correlated variables.
- Identify strongly negatively correlated variables.
- Find variables with weak or no correlation.
- Detect redundant features.
- Identify relationships that may be useful for predictive modeling.
- Detect possible multicollinearity between independent variables.

The correlation coefficient generally ranges from **-1 to +1**:

- **+1** → Strong positive correlation
- **0** → No linear correlation
- **-1** → Strong negative correlation

However, correlation only indicates association and **does not prove causation**.

Q4. Explain the purpose of histograms, box plots, and scatter plots in EDA.

Answer:

1. Histogram: A histogram shows the frequency distribution of a numerical variable. It helps identify the central tendency, spread, skewness, and distribution of the data.

2. Box Plot: A box plot summarizes the distribution of numerical data using the minimum, first quartile, median, third quartile, and maximum. It is especially useful for identifying **outliers**.

3. Scatter Plot: A scatter plot displays the relationship between two numerical variables. It helps identify correlations, trends, clusters, and unusual observations.

Q5. How can EDA help identify data quality issues before analysis?

Answer: EDA helps identify several data quality problems, including:

- Missing values.
- Duplicate records.
- Incorrect data types.
- Outliers.
- Inconsistent values or categories.
- Invalid or impossible values.
- Highly skewed distributions.
- Incorrectly entered data.
- Imbalanced categorical or target classes.

Functions such as `info()`, `describe()`, and `value_counts()` can be used to inspect the dataset and identify these problems.

Q6. Why is correlation important in predictive analytics? Can correlation imply causation?

Answer: Correlation is important because it helps identify relationships between variables. Highly correlated features may provide useful information for predicting a target variable. It can also help identify redundant features and multicollinearity.

However, **correlation does not imply causation**.

For example, if ice cream sales and the number of people swimming both increase during summer, they may be positively correlated. This does not mean that increased ice cream sales cause more people to swim. A third factor, such as **hot weather**, may influence both.

Therefore, correlation shows an association between variables but does not establish a cause-and-effect relationship.

Q7. Which visualization would you use to analyze categorical and numerical variables? Justify your choice.

Answer:

For **categorical variables**, I would use:

- Bar charts
- Count plots

These are useful for showing the frequency or number of observations in each category.

For **numerical variables**, I would use:

- Histograms
- Box plots
- Scatter plots

A histogram shows the distribution of numerical data, a box plot helps detect outliers and understand spread, and a scatter plot helps analyze the relationship between two numerical variables.

Therefore, the choice of visualization depends on the type of variable and the objective of the analysis.

Q8. What business insights can be derived from the Netflix (or Superstore/HR Analytics) dataset through EDA?

Answer:

Using EDA on a business dataset such as the Netflix, Superstore, or HR Analytics dataset, several useful insights can be obtained.

For example, in the **Superstore Sales Dataset**, we can analyze:

- Which product categories generate the highest sales.
- Which products or regions generate the highest profit.
- The relationship between sales and profit.

- The effect of discounts on profit.
- Which regions or customer segments perform best.
- Which products have low sales or profitability.

These insights can help businesses make better decisions regarding **products, pricing, discounts, customers, and regional strategies**.

Q9. How does EDA contribute to feature selection and model building?

Answer: EDA helps in feature selection by showing which variables are useful and which may be irrelevant or redundant.

It helps to:

- Identify important relationships between features and the target variable.
- Remove irrelevant features.
- Detect highly correlated or redundant features.
- Identify outliers that may affect the model.
- Understand feature distributions.
- Determine whether transformations or scaling are required.
- Identify missing values that need to be handled.

As a result, EDA can lead to cleaner data, better feature selection, and potentially more accurate and reliable machine learning models.
